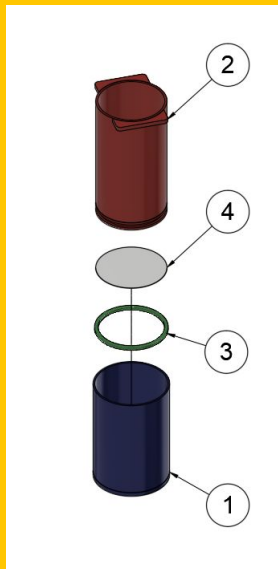


# A4: Engineering Validation Testing (EVT) Report

Team Bob the Builders: Will Castro, Jos Moresi, Li Arditi, and Sam Eppley





## The PressMate

A compact, single serving coffee-brewing device

# EVT Prototype Overview



Complete EVT Product



Vessel

Plunger

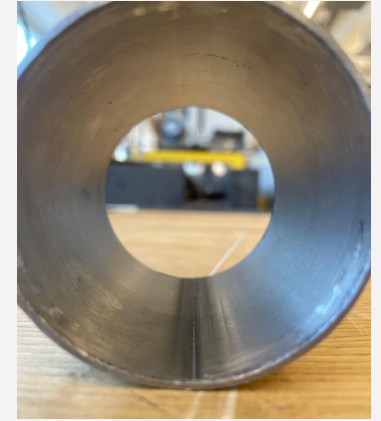
Sleeve

Lid

# EVT Prototype Overview

## Vessel

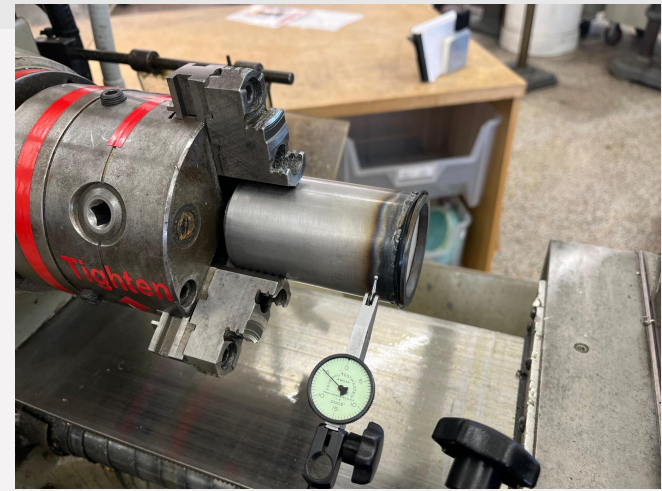
- Body:
  - Low-Carbon Steel Round Tube
  - 0.065" Wall Thickness, 2-1/4" OD
  - Cut to size via wire EDM
- Bottom:
  - Steel sheet
  - Thickness  $\frac{1}{8}$ "
  - Cut to size/shape via water jet
- De-burr and chamfer edges
- Bore interior surface to remove weld
- Polish exterior surface
- Weld bottom to body and turn to smooth



# EVT Prototype Overview

## Plunger

- Body:
  - Low-Carbon Steel Round Tube
  - 0.065" Wall Thickness, 2-1/2" OD
  - Cut to length with wire EDM
- Filter:
  - 316 Stainless Steel Wire Cloth
  - 1ft x 1ft Sheet 100 x 100 Mesh Size, 0.006" Opening Size
  - Cut to size with scissors
- O-ring:
  - Oil-Resistant Buna-N O-Ring
  - 0.118" Width, Dash Number 226
- Deburr and chamfer edges
- Weld tabs, bottom ring, and filter
- Turn to add o-ring notch



# EVT Prototype Overview

## Sleeve

- Two pieces of fabric sandwiching insulation layer
- Sewing machine and hand sewing to put together



## Lid

- Machined out of scrap  $\frac{3}{4}$ " HDPE on the CNC Mill
- Chamfers added for easy-of-use
- Deburred and sanded for aesthetics



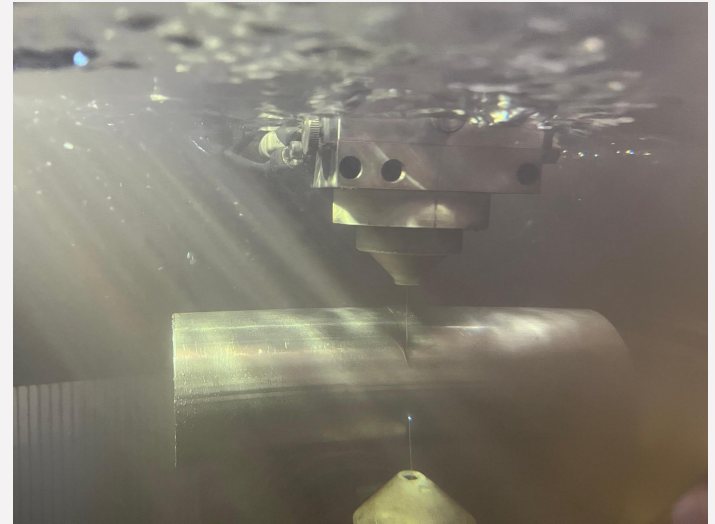
# Design Changes

- Square o-ring profile
  - prevent rolling when plunger is pressed through vessel
- Inclusion of sleeve
  - Insulates the vessel to prevent burning and keep drink warm
- Inclusion of lid
  - Protects the user's hand from steam or potential splashing.



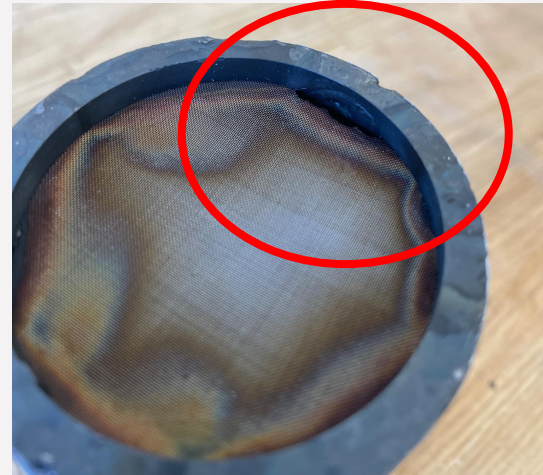
# Update to Processes

- Wire EDM to cut tubing to length (orig. Horizontal bandsaw)
  - Keeps the part in round
- Using a carbide cutting tool to machine through heat-treated steel.
  - Ran into problems with the tool grabbing
  - Had machined part fall out of the chuck



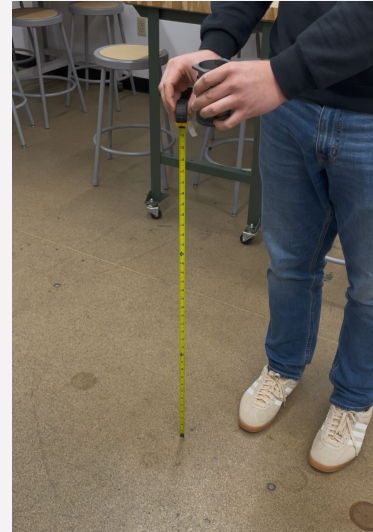
# Risks

- **Welding filter:** mesh melts extremely easily causing holes
- **Notch for o-ring:**
  - notch must be deep enough to securely seat o-ring but shallow enough to create proper seal with vessel.
  - Notch must in in-round with tube stock to ensure a proper seal.
- **Plunger-vessel seal:** the seal must be tight enough to prevent water from passing but loose enough to allow for minimal effort when pressing



# Next Steps

- Tooling
  - Create tooling for holding tubes during welding and aligning tabs
- Consider CNC lathe for boring and chamfers
- Conduct more extensive quality tests
- Order more materials
- Develop process to more efficiently produce PressMates
- Consider annealing vessels and plungers after welding



# Product Competition Update

## AeroPress Coffee Maker - Steel

Stainless Steel 3 in 1 Coffee Press

\$169.95 **NEW**

A new standard in coffee flavor. Crafted from double-wall stainless steel with a large capacity, our unique 3-in-1 brew method speeds up extraction for less bitterness and a smoother, more full-bodied cup.

Color: **Silver**

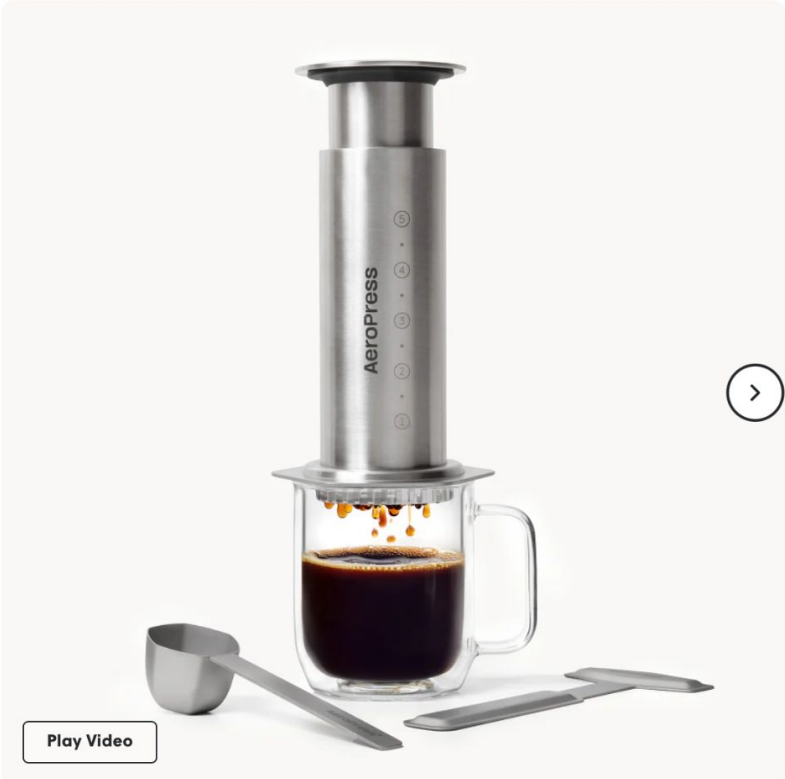


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